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EventPulse: Intelligent Event Registration and Ticketing System Using PHP and MySQL

1. Aim

To develop a web-based event registration and ticketing platform using PHP and MySQL that lets attendees choose exact seats on an interactive stadium map, protects passes against resale with a rotating cryptographic token, builds conference itineraries, applies squad discounts and records optional carbon-offset contributions.

2. Software Requirements

- PHP 8.3 (built-in development server, PDO extension)
- MySQL 8.0 (database eventpulse_db with 11 relational tables)
- Bootstrap 5.3 (responsive user interface)
- HTML5 Canvas 2D (stage perspective viewport)
- Web Audio API (gate-scan sound feedback)
- A modern web browser, with a laptop or smartphone camera for the optical QR scanner

3. Theory

EventPulse replaces flat ticket dropdowns and static barcodes with spatial seat selection, rotating passes and automated group perks. The system is built around five core features.

Interactive SVG seating map and stage perspective. Attendees pick seats from a tiered 96-node stadium grid covering VIP Sovereign, Premium Gold and Standard tiers. A Canvas 2D viewport calculates the stage distance (6 m to 26 m), the lateral angle offset (-36 to +36 degrees) and an acoustic clarity score for the chosen seat. To prevent double booking, MySQL row-level locks are used together with 5-minute checkout holds.

Anti-scalp rotating TOTP pass and audio gatekeeper. Each pass shows an 8-character token that is recomputed every 30 seconds using HMAC-SHA256, so a screenshot or a resold copy quickly becomes invalid. At the gate, organizers scan passes with the built-in

optical scanner on a phone or laptop camera. The Web Audio API plays a two-tone chime for a valid admission and a low saw-wave buzz for a duplicate check-in attempt.

AI event concierge and session matchmaker. The attendee enters a professional role (Engineer, Founder, Designer) and conference targets. The concierge builds a conflict-free chronological agenda, pairs the attendee with relevant networking cohorts, and exports an RFC-compliant .ics calendar file for Google Calendar, Apple iCal and Outlook.

Squad RSVP and group perks engine. A squad leader generates a shareable team code (for example SQ-TECH-4912) and invites colleagues. Applying the code at checkout gives a 15% to 20% discount on all passes and generates coordinated team delegation passes.

Demand surge and carbon-offset EcoBadge. Live telemetry tracks hourly ticket velocity and triggers early-bird countdowns. An optional $5 contribution funds 2 native trees and offsets 50 kg of CO2, and attaches an EcoGuardian badge and certificate hash to the pass.

Database design. The schema is normalized into 11 relational tables. The main ones are shown below.

Table	Role and Responsibility
users	Authentication, bcrypt hashes, role-based access (admin, organizer, attendee)
events	Event profile, category, venue name, address, dates, seating configuration
ticket_tiers	Pricing tiers (VIP, Premium, Standard), capacity, perks JSON
seats	Seat coordinates, rows, status (available, reserved, booked)
bookings	Order records, references, subtotals, discounts, net amount
ticket_items	Issued passes, attendee info, SHA-256 hash, check-in status
squads	Team delegation records, referral codes, target sizes, member progress
carbon_offsets	Trees funded, kg CO2 offset, certificate hashes

Security. Passwords use PHP password_hash() with BCrypt (work factor 10). All queries use PDO prepared statements with bound parameters, which prevents SQL injection. Output is escaped with htmlspecialchars() against XSS, and state-changing requests carry signed session CSRF tokens.

4. Algorithm

1. Run the setup script php config/setup.php to create eventpulse_db and load the seed data.
2. Start the development server using php -S localhost:8000 (or run_server.bat).
3. Open the site in a browser and register or log in. The role (admin, organizer or attendee) decides which pages are available.
4. Browse events and open an event to view its stadium seating map.
5. Select seats. The Canvas viewport computes stage distance, angle offset and acoustic clarity for the chosen seat.
6. Place a 5-minute checkout hold on the selected seats using MySQL row-level locks, so no other user can book them.
7. Optionally enter a squad code. If valid, apply the 15% to 20% discount and create team delegation passes.
8. Optionally add the $5 carbon-offset contribution and record the trees funded, CO2 offset and certificate hash.
9. Confirm the booking. Save it in bookings, issue passes in ticket_items with a SHA-256 hash, and mark the seats as booked.
10. Show the pass with its 8-character token, recomputed every 30 seconds with HMAC-SHA256.
11. At the venue gate, the organizer scans the pass with the optical scanner and the token is validated.
12. Play the two-tone chime and mark the pass checked in if it is valid. Play the low saw-wave buzz if the pass was already used.
13. For conferences, take the attendee's role and goals, generate a conflict-free agenda and matching cohorts, and export a .ics calendar file.
14. Update the hourly ticket-velocity telemetry and show early-bird countdowns when demand surges.
15. Release expired holds, then stop.

5. Program

The project is a PHP web application. Run config/setup.php once to create the database, then launch the server and open http://localhost:8000.

Note: only the project report was provided to me, not the source files. Paste each file's code under its heading below.

5.1 config/setup.php (database setup and seeder)

[Paste code here]

5.2 Seat selection and checkout hold (row-level locking)

[Paste code here]

5.3 Rotating TOTP pass generation and validation (HMAC-SHA256)

[Paste code here]

5.4 Gate scanner and audio feedback (Web Audio API)

[Paste code here]

5.5 AI concierge and .ics export

[Paste code here]

5.6 Squad codes and carbon-offset module

[Paste code here]

6. Output and Screenshot

Run the server, exercise each feature, and insert one screenshot of the result per row below. Fill in the Observed column from what the browser shows.

Feature	Action	Expected Result	Observed
Seating map	Select a seat in the 96-node grid	Distance, angle and acoustic score shown in the Canvas viewport	

Feature	Action	Expected Result	Observed
Concurrency	Try to book a held seat from a second session	Seat is locked for 5 minutes	
Squad discount	Apply a valid squad code at checkout	15% to 20% discount applied	
Carbon offset	Add the $5 contribution	EcoGuardian badge and certificate hash on the pass	
Rotating pass	Watch the pass for 30 seconds	8-character token changes	
Valid scan	Scan a valid pass at the gate	Two-tone chime, check-in recorded	
Duplicate scan	Scan the same pass again	Low saw-wave buzz, entry rejected	
AI concierge	Enter role and targets, then export	Conflict-free agenda and .ics file	

Demo accounts: admin@eventpulse.io / admin123, organizer@eventpulse.io / organizer123, attendee@eventpulse.io / attendee123.

7. Result

Thus, EventPulse, a web-based event registration and ticketing system, was developed using PHP, MySQL and Bootstrap. It provides interactive seat selection with stage perspective simulation, anti-scalp rotating TOTP passes with audio-verified gate admission, an AI itinerary concierge, squad group discounts and a carbon-offset EcoBadge, all backed by an 11-table normalized database.

Note: Security measures include BCrypt password hashing, PDO prepared statements, output escaping and CSRF tokens. The observed results in Section 6 depend on the demo data and on camera and browser permissions for the scanner.